Ogbonna Arinze

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SUMMARY

Versatile Senior Fullstack Engineer with 6+ years of experience delivering high-impact web applications across the entire stack, from intuitive UX and responsive frontends to optimized databases and scalable backend services. Proven expertise in TypeScript, React, Node.js, PostgreSQL, and GraphQL, with hands-on experience in Docker, automated testing (Jest, Cypress, Playwright), and CI/CD automation. Adept at leading end-to-end feature development, architecting production systems, and integrating AI capabilities into enterprise applications. Passionate about long-term product development, engineering quality, and building software that drives measurable business outcomes.

CORE COMPETENCIES

Fullstack Development: React, TypeScript, Node.js, Next.js, PostgreSQL, GraphQL, Hasura, TailwindCSS, Radix UI.

Backend & Data: PostgreSQL, Prisma/Drizzle ORM, REST APIs, GraphQL APIs, JWT Authentication, Role-Based Access Control (RBAC), Redis.

Frontend & UX Engineering: Design Systems, Accessibility (WCAG), Responsive Design, SSR/SSG, Figma-to-Code Workflows, Core Web Vitals.

Testing & Quality: Jest, Cypress, Playwright, TDD/BDD, CI-Driven QA, Allure Reports, Storybook.

DevOps & Infrastructure: Docker, Kubernetes, GitHub Actions, CI/CD Pipelines, AWS (CloudFront, S3), Vercel, Edge Rendering.

AI Integration: LLM Integration, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vercel AI SDK, LangChain.

Monitoring & Observability: Sentry, Datadog RUM, LogRocket, Lighthouse, Chrome DevTools, React Profiler.

Leadership & Methodologies: Technical Direction, Architecture Reviews, Mentorship, Code Reviews, Agile/Scrum/Kanban.

WORK EXPERIENCE

Roche Genentech

San Francisco, CA (Remote)

Senior Fullstack Engineer

May 2023 - Present

- **Architected** the fullstack platform for a clinical trial data system, designing a reusable **React/TypeScript** component library with **Radix UI** on the frontend and a **GraphQL** API layer backed by **PostgreSQL**, ensuring data integrity, accessibility, and developer efficiency across multiple products.
- **Spearheaded** the end-to-end migration of a legacy **Vue.js** application into a modern **React-based Chrome extension**, delivering changes from database schema updates through API modifications to frontend UI, optimizing clinical site integration workflows and driving team adoption.
- **Optimized** application performance across the full stack by implementing lazy loading, code splitting, and dynamic imports on the frontend while tuning **PostgreSQL** queries and leveraging CDN asset delivery, reducing page load times significantly and improving **Core Web Vitals** scores using **Lighthouse** and **Chrome DevTools**.
- **Engineered** a robust authentication and authorization system using **JWT** tokens with HTTP-only cookies and **role-based access control (RBAC)** in a **Next.js/TypeScript** application, collaborating with backend teams to secure API endpoints and enforce permission-based UI rendering.
- **Scaled** state management in a large **Next.js** application by integrating **Redux** for complex client state and **React Query** for cached **GraphQL** API data, streamlining data flow, eliminating redundant re-renders, and accelerating developer onboarding through consistent, maintainable patterns.

- **Increased** release reliability by integrating **Playwright** and **Cypress** end-to-end tests into the **CI/CD pipeline**, automating test reporting with **Allure Reports** on **GitHub Pages**, and configuring Slack notifications for real-time visibility into build quality and test outcomes.
- **Designed** and implemented **GraphQL** APIs to support complex data queries for clinical trial dashboards, leveraging **PostgreSQL** for relational data modeling and optimizing query performance through indexing, query plan analysis, and efficient resolver patterns.
- **Containerized** frontend and backend services using **Docker**, establishing reproducible development environments and streamlining deployment workflows across staging and production, reducing environment-related configuration issues by 60%.
- **Built** real-time data synchronization features using **WebSockets** and server-sent events, enabling live updates on clinical dashboards and improving operational responsiveness for end users monitoring active trial data.
- **Led** cross-functional collaboration with product, design, and backend teams to deliver feature increments from UX wireframes through database schema design to production deployment, driving a culture of end-to-end ownership and rapid iteration.
- **Implemented** comprehensive monitoring and observability using **Sentry** for error tracking, **Datadog RUM** for performance monitoring, and **LogRocket** for session replay, proactively detecting and resolving issues before customer impact.

DESCASIO LTD Lagos, Nigeria

Staff Fullstack Engineer *May 2021 - Apr 2022*

- **Owned** the fullstack architecture for a data-heavy enterprise **React** dashboard with a **Node.js** backend and **PostgreSQL** database, implementing virtualization, efficient state management, and **GraphQL** subscriptions to handle large datasets seamlessly across the stack.
- **Optimized** frontend performance by identifying and resolving render bottlenecks using **React Profiler** and **Lighthouse Audits**, reducing bundle size by 30-40% through **rollup-plugin-visualizer** and **webpack-bundle-analyzer**, achieving a 35% faster load time and smoother runtime experience across devices.
- **Designed** and implemented RESTful and **GraphQL** APIs in **Node.js** with **PostgreSQL**, building efficient data access layers using **Prisma ORM**, implementing pagination, filtering, and aggregation for complex enterprise reporting dashboards.
- **Established** engineering standards for code quality, testing, and deployment using **ESLint**, **Prettier**, **Husky**, **Lint-staged**, **Jest**, and **Cypress**, integrating automated test suites into **GitHub Actions** CI/CD pipelines, significantly reducing production errors.
- **Built** and maintained **Docker**-based development and deployment environments, standardizing local setups across the engineering team and enabling consistent, reproducible deployments through containerized CI/CD workflows.
- **Architected** a component design system with **TailwindCSS** that ensured visual consistency across the platform, implementing responsive layouts, theming, and **accessibility standards (WCAG)** for enterprise clients.
- **Implemented** event-driven data processing workflows using **Node.js** and **PostgreSQL** triggers, enabling real-time data updates, audit logging, and complex workflow orchestration for critical business operations.
- **Mentored** junior engineers on fullstack development best practices, conducting code reviews, architecture sessions, and pair programming on complex features, fostering a culture of high-quality, maintainable code and consistent design.
- **Led** the migration of legacy frontend code to **TypeScript**, improving type safety, reducing runtime errors by 25%, and enhancing developer productivity through better IDE support and self-documenting code.
- **Integrated** AI-powered recommendation features into the enterprise dashboard using LLM APIs and **prompt engineering** patterns, implementing retrieval-augmented generation (RAG) to surface contextual insights for end users.

EDUCATION

University of Nigeria Nsukka, Enugu

BSc Computer Science, 4.1 CGPA *Graduation Date: Sep 2019*